

JAI LI

(301) 283-8308 | Tommyli874@gmail.com | linkedin.com/in/jai-li-va | github.com/lijw07 | lijw07.github.io/portfolio/

EDUCATION

Georgia Institute of Technology | Atlanta, GA

M.S. Computer Science, OMSCS

Jan 2025 – Present

GPA: 4.0

Virginia Commonwealth University | Richmond, VA

B.S. Computer Science, College of Engineering

Jan 2020 – May 2022

TECHNICAL SKILLS

Languages | Java, Python, JavaScript/TypeScript, C#, SQL, Bash, GDScript

Frameworks | Spring Boot, React, ASP.NET Core, Entity Framework, REST, JPA, Flask, Unity, Godot

Build & CI/CD | Maven, Gradle, NPM, Jenkins, GitLab CI, GitHub Actions, Docker, Kubernetes

Cloud & DevOps | AWS (S3, RDS, Lambda, Chalice, SNS/SQS), GCP, Linux, Datadog, Postman, Swagger/OpenAPI, Ollama

Databases | MySQL, PostgreSQL, Oracle, SQL Server, MongoDB, DynamoDB

Testing & Security | JUnit, xUnit, Mockito, OAuth 2.0, JWT, OWASP Top 10, secure coding, SAST

WORK EXPERIENCE

U.S. Patent and Trademark Office (USPTO) | Patent Examiner

Oct 2025 – Present

- Examine U.S. patent applications in Technology Center 2600 (Art Units 2613, 2859), covering optical/display/imaging and digital communications technologies.
- Conduct prior art searches across U.S., foreign, and non-patent literature databases; apply Broadest Reasonable Interpretation (BRI) to map claim limitations against references.
- Draft non-final and final office actions including §101, §102, §103, and §112 rejections in compliance with MPEP guidelines and KSR/Alice/Mayo case law.

Brightspot | Software Engineer

Jun 2022 – Jun 2024

- Developed RESTful Java/Spring microservices and servlet-based APIs for a headless CMS platform, building Maven-assembled artifacts deployed via GitLab CI/CD pipelines.
- Maintained Kubernetes clusters with Docker-containerized services that auto-scaled environments and absorbed traffic surges with zero downtime; led Agile/Scrum architectural design across 2-week sprints.
- Wrote JUnit and Mockito unit/integration tests and applied OWASP-aligned secure coding practices (input validation, authz checks, dependency scanning) to harden services against regressions.

PROJECTS

Slop – AI-Assisted Prior Art Search Tool

Jan 2026 – Present

- Built an AI-assisted research tool that surfaces relevant U.S. and foreign prior art across publicly available patent corpora given a target application's claims and specification.
- Developed a Spring Boot backend (MVC, Java 17, Maven) orchestrating a locally-hosted LLM (Qwen2.5 via Ollama) over LAN, with a Google Patents scraping subsystem feeding a ReAct-style iterative search and refinement loop.
- Designed a card-based front end (PatentInfoCard, ClaimsCard, SpecificationCard) for browsing candidate references with all validation and processing handled in the backend service layer.
- Implemented MongoDB persistence, streaming response endpoints, JUnit test coverage, and prompt-level guardrails to reject malformed or unreliable LLM outputs.

Centralized Application Management System (CAMS)

Jun 2025 – Present

- Built a multi-tenant full-stack web application with React, ASP.NET Core, and Entity Framework, exposing secured REST APIs for unified cross-system data management.
- Integrated AWS RDS, GCP-hosted MySQL, PostgreSQL, Oracle, SQL Server, MongoDB, and DynamoDB with a migration layer supporting up to 20 concurrent data sources per session.
- Implemented OAuth 2.0 / JWT authentication and role-based access control with audit logging for all CRUD operations against connected data sources.
- Wrote xUnit tests, containerized services with Docker, and deployed to Kubernetes via GitHub Actions CI/CD.

Paws and Hooves – 3D Indie Adventure Game

Jun 2024 – Dec 2024

- Built a 3D puzzle-adventure game in Unity (C#) featuring four playable animal characters (Cow, Donkey, Chick, Chicken), each with distinct abilities driving puzzle and traversal mechanics.
- Implemented NPC behavior trees and pathfinding with Unity NavMesh and the Animator state machine, simulating farm life, traffic, and environmental interactions.
- Engineered real-time gameplay systems and a full UI: day/night lighting cycle, enemy AI, map navigation, ability cooldowns, inventory, and a settings panel with persistent player preferences.

Tower Defense Game – Indie Portfolio Project

Jan 2026 – Present

- Built a real-time tower defense game in Godot 4 / GDScript with wave-based spawning driven by Resource-based WaveData and EnemyData for data-driven content authoring.
- Implemented enemy pathing (Path2D / PathFollow2D), Area2D-based castle collision detection with signal-driven event handling, upgradeable tower mechanics, and save/load UI systems.